

STAT 5700 — Quiz 4

Date: March 11, 2026

SOLUTIONS

Problem 1 (0.5pt each) For each of the following scenarios, fill in the blank with the distribution that is most appropriate to model the random variable X .

Geometric A product team is testing a new mobile app feature. Based on past data, 15% of users who see the feature click on it. Let X be the number of users who see the feature until the first user clicks it.

Hypergeometric A university randomly selects 3 students from a class of 20 to serve on a student advisory panel. In the class, 9 students are statistics majors. Let X be the number of statistics majors among the selected students.

Poisson A financial fraud detection system flags suspicious transactions. On average, the system flags 4 suspicious transactions per day. Let X be the number of suspicious transactions flagged in a one-week period.

Binomial A manufacturing company produces microchips. Historically, 3% of chips are defective. A quality control inspector randomly selects 10 chips from the production line. Let X be the number of defective chips in the sample.

[FLIP TO BACK]

Problem 2 (2pts) A researcher needs a sample size of 35 for their survey. Suppose people only respond with 30% probability.

- a) On average, how many people does the researcher need to contact to ensure 35 people respond?

$$Y \sim \text{NegBinomial}(r = 35, p = 0.3)$$
$$E(Y) = \frac{r}{p} = \frac{35}{0.3} \approx 117$$

- b) What is the probability that the researcher will need to contact at most 100 people? Write out the formula (with appropriate numbers plugged in) - you do not need to calculate the result.

$$P(Y \leq 100) = \sum_{y=35}^{100} \binom{y-1}{34} (0.3)^{35} (0.7)^{y-35}$$

TRUE/FALSE (0.5pt each)

3. **FALSE** If $Y \sim \text{Poisson}(\lambda)$ and $V(Y) = 4$, then $E(Y^2) = 16$.
4. **FALSE** The geometric distribution is a special case of the hypergeometric distribution, with $r = 1$